Gatekeeper — One-Page Project & Impact Flow Diagram

IUB Innovation to Impact (i2i) Award Program | Phase 1, 2026

> *This diagram illustrates the user-centered flow of Gatekeeper—from the core problem faced by university event organizers to the delivered outputs and long-term societal and commercial impact routes.*

Visual Impact Diagram (One View)

```
flowchart TD
    subgraph S1["1. PROBLEM / NEED"]
        P1["Manual Paper Check-in Lists"]
        P2["Severe Gate Queues (45s per entry)"]
        P3["Ticket Forgery & Unverified Certificates"]
    end

    subgraph S2["2. TARGET USER / CLIENT"]
        U1["Primary: IUB Division of Student Affairs & St
        U2["Secondary: External Event Agencies & Universi
    end

    subgraph S3["3. EXISTING OUTPUT (Gatekeeper Platform)
        O1["Mobile Camera QR Scanner (<2s scan)"]
        O2["Self-Entry Kiosk Terminal"]
        O3["Real-time Gate Capacity Analytics"]
        O4["Automated PDF Certificate Generator & Verifie
    end

    subgraph S4["4. INPUTS NEEDED"]
```

```
    I1["Official DSA Campus Deployment Mandate"]
    I2["Cloud Database & SMS Gateway Hosting Support"
    I3["i2i Mentorship for B2B SaaS Commercialization
end

subgraph S5["5. PROCESS / PATHWAY"]
    W1["Phase 1: Registration & Encrypted QR Ticket I
    W2["Phase 2: Instant Gate Access & Real-Time Capa
    W3["Phase 3: Post-Event Automated Certificate Gen
    W4["Phase 4: Public Verification via Authenticity
end

subgraph S6["6. OUTPUTS & IMPACT ROUTE"]
    R1["Institutional Impact: Zero Gate Queues & 100%
    R2["Commercial Route: B2B Event Management SaaS f
    R3["Employability: Production-grade Engineering P
end

S1 --> S2
S2 --> S3
S3 --> I4["Support Requested"] --> S4
S4 --> S5
S5 --> S6
```

Flow Component Breakdown

Component	Key Details
1. Problem / Need	Manual event access at campus venues causes major entry queues (up to 45 seconds per person), risk of crowd safety hazards, vulnerability to ticket duplication, and days of post-event administrative burden typing attendee lists for certificate generation.

2. Target User / Client	**Primary:** IUB Division of Student Affairs (DSA), IUB Student Clubs (IUB Photography Club, IEEE, FabLab, Club Forum). **Secondary:** External private universities, corporate convention centers, and event management agencies across Dhaka.
3. Existing Output	A fully built, tested, production-grade web application (`iubpc-gatekeeper`) featuring instant browser-based QR scanning, self-service kiosk check-in, real-time analytics dashboards, and automated PDF certificate issuance with tamper-proof verification.
4. Inputs Needed	Institutional adoption sanction from IUB DSA, minor cloud hosting budget support (Supabase Pro & SMS ticket API), and business/IP mentorship through the i2i incubation track.
5. Process / Pathway	1. **Organizer Setup:** Create event, set registration fields, custom pass layout. 2. **Attendee Registration:** Guests register online and receive encrypted QR passes. 3. **Gate Scanning:** Volunteers scan passes via phone camera; system verifies in <2 seconds. 4. **Instant Issuance:** Attendees receive verifiable digital certificates immediately after event completion.
6. Outputs & Impact Route	**Campus Scale:** Eradicates event entry queues, enforces zero ticket fraud, and automates 100% of certificate workflows at IUB. **Commercial Scale:** Expands into a subscription SaaS platform for event organizers nationwide. **Career Impact:** Equips student lead & developers with scalable, real-world deployment experience.